

EDUCATION

UNIVERSITY OF COLORADO BOULDER

2019
MS in Aerospace Engineering
Research on numerical optimization

CORNELL UNIVERSITY

2018
BS in Electrical and Computer Engineering
Minor: Aerospace Eng. & Archaeology

SOFTWARE

Proficient

C/C++ • Python • Git
Matlab • $\text{\LaTeX}$

Design:

Fusion360 • EAGLE

HARDWARE

Circuit Design and PCB Layout
Embedded Systems Development
Mechanical Design
3D Printing
Machining

AWARDS

2018: Matthew Isakowitz Fellow
2017: Winner Caltech Space Challenge
MakeMIT Amazon Prize

INDUSTRY

SPACEX | 2020 - PRESENT**GUIDANCE, NAVIGATION, AND CONTROL ENGINEER II****STARSHIP AND LUNAR STARSHIP**

- Responsible Engineer for Starship Flight 9
- Contribution to all Starship missions: simulation, Monte Carlo, algorithm development, Raptor propulsion modeling, configuration automation, flip and landing burn design work
- Responsible Engineer for Lunar Landing: designed and implemented the guidance, control, and allocation algorithm stack from cislunar space, down to the surface of the moon. Helped build a modular simulation infrastructure for iterative feature development.
- On-board real-time trajectory optimization routines for rocket landing with scratch-built, robust solvers
- Offline sequential convex programming techniques for mission and trajectory design
- Led GNC for \$150M+ in NASA milestone products
- Developed landing manual control schema with NASA Crew Office, held demonstrations and maintained working relationships with

STARSHIELD

- Cradle-to-grave development of new program GNC: wrote and flew over a dozen new algorithms. Demonstrated successful fault detection, recovery, on-orbit operations, and completed full data review cycle. Worked on a small team of four.
- Implemented attitude control, orbit maintenance/maneuvering, state machine design and transition algorithms, device deployment schema and sequences, peripheral device control and fault detection, isolation, and recovery
- Sole maintainer of orbit maneuvering and collision avoidance systems for new satellite constellation. A new team was created for my replacement

SPACEX | INTERNSHIP/ASSOCIATE POSITIONS | SUMMERS 2015/16/18/19

- Four internships across three sites, working on Falcon, Dragon, and Starlink
- Launch instrumentation on Pad 39A, Dragon RF reradiation system for TDRS checkouts, Starlink Attitude Determination and Control avionics, Gateway antenna actuation circuit design

CORNELL SPACE SYSTEM DESIGN STUDIO AUG 2014 – NOV 2017 | ITHACA, NY

- Avionics Lead on Artificial Gravity Cubesat under Dr Daniel Selva; complete in-house avionics design to demonstrate controlled artificial gravity.
- Avionics Lead on the Violet Nanosatellite Project under Dr. Mason Peck.

PERSONAL PROJECTS

THE NOTFORFLIGHT LANDER PROJECT | '20 - PRESENT | WWW.NOTFORFLIGHT.COM

- A pressure-fed, liquid bipropellant, vertical-takeoff-vertical-landing testbed for novel guidance algorithm routines.
- Scratch built film-cooled liquid rocket engine, electric Thrust Vector Control actuators, throttle valve, avionics, flight software and control stack in C++
- Status: fluid systems and flight software are being tested, leading to goal of static fire summer 2026

PUBLICATIONS

JAN '21 – P. Lysandrou, RD Braun, A 6-DoF Successive Convexification Powered Descent Guidance Implementation using Modified Rodrigues Parameters, AIAA Scitech 2021 Forum (Conference)